Daniil Lysenko

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ASTROPHYSICS CV
St. Petersburg, Russia

Research interests: Computational/theoretical astrophysics; quasar absorption systems; ISM/IGM; neutron stars; plasma astrophysics.

Education

Alferov University (St. Petersburg Academic University) *Expected June 2027*
B.Sc. in Physics, Astrophysics Track **GPA: 4.82/5.00**
Current coursework: Statistical Physics; Nuclear Astrophysics; Quantum Field Theory; Statistical Analysis of Astrophysical Data; X-ray and Gamma-ray Astrophysics.

Novosibirsk State University — Physics coursework, 2023–2024; first-year GPA: 4.90/5.00 (transferred in 2024).

Research Experience

Ioffe Institute, Theoretical Astrophysics Department *Oct. 2024–Present*
Research Assistant

Quasar 4C 04.81: partial covering, absorber structure, and disk inference *Dr. Sergei Balashev*
- Analyzed VLT/UVES and X-Shooter spectra ($z = 2.58$), identifying 31 transitions from 12 elements, detecting partial covering at 10σ, constraining cloud size (> 0.019 pc), and estimating continuum-slope disk exponent $\beta = 0.571 \pm 0.036$.
- Extended SPECTRO with partial/inhomogeneous absorber models and AIC/BIC comparisons; a joint 830-pixel Fe II/Fe II* fit favored by-system covering over the best common solution ($\Delta\mathrm{BIC} \approx 122$). Fe II excitation dependence is established; disk chromaticity remains under investigation.

GRB light-curve inference with random Fourier features *Dr. Dmitry Svinkin*
- Recovered and repaired a Stan/Poisson random-Fourier-feature inference model from `pyipn/nazgul`; validated single, double, and faint synthetic GRB pulses (peak-scale NRMSE ≈3–7.2%).

Neutron-star structure and magnetic-field evolution *Dr. Mikhail Gusakov*
- Implemented six TOV solvers (BSk24: $M_{\max} = 2.28\,M_\odot$) and modeled magnetic winding, Ohmic diffusion, and torsional Maxwell feedback with kinematic and reduced-order calculations.

ITMO University, Metamaterials Laboratory *Mar. 2026–Present*
- Designed 5 GHz Huygens meta-atoms (43 high-transmission phase states) and 377-element metalens/deflector models using COMSOL, FDTD, and LightPipes; simulated deflection near a 20° target.

Special Astrophysical Observatory (SAO RAS) *July 2026*
- Observational practicum: RATAN-600/FITS data reduction, VizieR/TOPCAT/Aladin source analysis, and a 6-m BTA observing session.

Budker Institute of Nuclear Physics *July–Aug. 2024*
- Ran LCODE particle-in-cell wakefield simulations on HPC systems and analyzed beam/field evolution.

Selected Presentations

"Accretion Disk Models and Quasar Absorption Systems" (2026); **"Spectral Methods in Relativistic Astrophysics: Modeling Magnetic Field Dynamics in Rotating Neutron Stars"** (2026). Both contributed student-conference talks.

Technical Skills & Languages

Programming/computing: Python (NumPy, SciPy, Astropy, pandas), Stan/CmdStanPy, C/C++, Bash, Linux, Git, LaTeX; Dedalus, SPECTRO, COMSOL, LCODE, MPI.
Methods: Spectroscopy/Voigt fits; numerical PDEs and TOV solvers; Monte Carlo/Bayesian inference; FITS and astronomical-catalog analysis.
Languages: Russian (native); English (C2, TOEFL iBT 6/6; comparable 118/120).

Teaching, Honors & Leadership

Pro Math Online — Mathematics/physics tutor for international students in English, Russian, and bilingual formats (2026–Present); also Tetrika (2026) and freelance tutoring (2025–Present), 45+ students across roles.
Honors: EducationUSA Opportunity Funds grant (2022); academic scholarship (2023–Present); federal secondary-school gold medal, highest honors, 5.0/5.0 (2023).
Service/activities: Alferov group representative (2026–Present); EducationUSA Competitive College Club organizing committee (2021–2023); ICPC Northern Eurasia regional competitive-programming participant (2025).